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RESEARCH ARTICLE

Performance optimization of thermal integrated-Carnot battery for waste heat utilization in industrial integrated energy systems

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HIGHLIGHTS

- A set of dynamic performance indicators for Carnot battery was established.
- A sensitivity analysis was conducted on key design parameters of the Carnot battery.
- A multi-operating condition set was performed via multi-variable sampling.
- The effects of design parameters and different organic Rankine cycle (ORC) working fluids on Carnot battery performance were systematically analyzed.

Keywords:
Carnot battery
Thermodynamic performance
Integrated energy systems
Modeling
Waste heat recovery

ABSTRACT

Thermally integrated Carnot battery (TI-CB) systems offer unique advantages for industrial waste heat recovery, but their performance under fluctuating, off-design conditions remains poorly understood. To address this gap, this study proposes a quasi-dynamic mathematical model with solution methodologies applicable to both design and off-design operating conditions. A dynamic evaluation framework is also developed to account for the temporal mismatch between energy storage and release processes. A multi-operating-condition set constructed via multivariable sampling is used to enable systematic analysis of key design parameters under both design and off-design conditions. The results reveal that heat source utilization parameters and heat pump temperature rise are dominant factors affecting TI-CB performance, while off-design analysis shows that ORC mass flow rate variations have a more significant impact on system performance than heat pump fluctuations. Due to irreversible heat losses, an increase in the heat source temperature difference leads to a decrease in round-trip efficiency (η_{rt}) from 62.6% to 45.8%, while η_{orc} and η_{ex} also exhibit downward trends. A higher temperature lift in the heat pump results a decrease in the mean COP from 7.6 to 4.8, whereas η_{orc} increases from 7.0% to 10.2%. Among working fluids evaluated, R1336mzz(Z) demonstrates superior performance but exhibits nonlinear behavior, while R1233zd(E) provides optimal stability across operating ranges, making it suitable for practical engineering applications.

1 Introduction

In 2024, integrated industrial energy systems accounted for nearly 40% of global electricity demand growth, while industrial carbon emissions increased by approximately 1% [1]. Under the dual pressures of global climate change and energy security, accelerating the transformation of industrial energy structures has become a global consensus [2]. Current energy conversion and storage systems, including electric boilers and gas-steam combined cycle plants, remain heavily dependent on fossil-based technologies and exhibit significant limitations in multi-energy flow coupling and efficient energy storage [3,4]. In response, various novel long-duration energy storage technologies have emerged to address renewable energy integration challenges. Compressed air energy storage (CAES) systems store

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energy through electricity-driven compression [5,6] but face efficiency limitations due to thermal management constraints and site-specific geological requirements [7].

In contrast, the Carnot battery—an emerging thermal energy storage technology—stores and releases energy through heat–mass transfer and reversible heat–work conversion. Dumont et al. [8] first systematically provided a clear definition of the Carnot battery: input electrical energy is used to establish a temperature difference between cold and hot thermal reservoirs, thereby enabling electricity storage in the form of thermal energy. Carnot batteries offer unique advantages in terms of site flexibility, power–energy decoupling, and multi-energy flow integration. Their operating principle is grounded in the second law of thermodynamics [9], enabling efficient energy storage and release through heat exchange between high- and low-temperature reservoirs. Compared with conventional electrochemical energy storage systems, Carnot batteries exhibit higher energy density, longer cycle life, improved environmental compatibility, and enhanced safety, making them particularly suitable for energy-intensive regions such as industrial parks. Consequently, Carnot batteries hold strong application potential in integrated industrial energy systems, providing an effective solution to the dual challenges of renewable energy integration and industrial energy demand.

Research on Carnot batteries began in the early 20th century. Marguere [10] filed the first preliminary patent on energy storage technology utilizing a heat pump cycle. Based on the principle of energy storage via the reverse Carnot cycle, this system established a multi-energy conversion framework of "electricity–heat–potential," thereby initiating the exploration of heat pump energy storage technology. Carnot batteries based on the organic Rankine cycle (ORC) and its derivatives demonstrate distinct advantages in industrial waste heat recovery. Research on ORC-based Carnot batteries has progressed along two main directions: system configuration optimization and working fluid selection. In terms of system design, Eppinger et al. [11] developed a coupled ORC and two-tank thermal storage model, identifying R245fa as the optimal working fluid with an energy storage density of 35 kWh/m^3. Building on this framework, Li et al. [12] optimized working fluid combinations for ORC Carnot batteries, achieving a round-trip efficiency of 109% at a heat source temperature of 105 °C using R1336mzz(Z) and R1234ze(Z). Their results indicate that optimized working fluid selection can enhance system round-trip efficiency by 20% to 125%.

Beyond Rankine-type Carnot batteries, other cycle configurations have also achieved significant progress. Zamengo et al. [13] developed a Carnot battery employing a supercritical CO_2 Brayton cycle with $CaO/Ca(OH)_2$ thermochemical storage, enabling high-temperature heat storage at 650 °C. However, its applicability in waste heat recovery remains constrained by system complexity and high material costs. Albert et al. [14] introduced cascaded phase-change thermal storage into a Brayton cycle, enhancing thermal storage efficiency within the 300–500 °C range to 92%, although the system integration complexity was considerably higher than that of Rankine-type Carnot batteries. Current research consensus indicates that Carnot batteries based on ORC possess significant advantages in industrial waste heat recovery due to their working-fluid diversity and adaptability to medium- and low-temperature ranges. Such systems are commonly referred to as the thermally integrated Carnot batteries (TI-CBs), reinforcing their position as the most practical and mature technology for industrial applications, combining technical effectiveness with operational simplicity and cost-effectiveness.

With TI-CB established as an optimal configuration, accurate modeling methodologies become essential for system optimization. Research on Carnot battery modeling has primarily evolved along two methodological frameworks: steady-state and dynamic modeling. These approaches differ significantly in model architecture, governing equations, and solution strategies. In steady-state modeling, Miao et al. [15] developed an integrated model based on the first law of thermodynamics, coupling compressor power, expansion work, and heat exchanger exergy losses through iterative optimization algorithms. Zhao et al. [16] extended traditional energy analysis by incorporating exergy-based methods to identify irreversible loss sources in the Joule–Brayton Carnot batteries. Liu et al. [17] proposed a multi-physics coupled approach for supercritical CO_2 cycles, solving nonlinear thermophysical property equations alongside Brayton cycle state equations.

Dynamic modeling frameworks focus on transient processes and time domain response characteristics. Chiapperi et al. [18] evaluated bidirectional turbomachinery performance in forward and reverse Brayton cycles, while Pecchini et al. investigated off-design operation, revealing that thermal storage temperature significantly affects discharge performance [19]. Zhao et al. [20] developed a stratified model for shell-and-tube latent heat storage using stage-wise energy balance and heat transfer correlations with storage structure optimization guided by entropy generation minimization. Tian et al. [21] proposed a finite-time thermodynamics framework incorporating process duration as an optimization variable, while Zhang et al. [22] employed higher-order differential equations system via fourthorder Runge–Kutta methods to analyze dynamic exergy efficiency evolution. He et al. [23] developed a multi-component coordinated dynamic numerical model that characterizes inertial effects during compression and expansion using rotor dynamic equations and integrates a proportional-integral-derivative (PID) control algorithm to regulate rotational speed. Wang et al. [24] proposed a transient analysis method coupling dynamics, heat transfer, and thermodynamics to investigate the energy storage characteristics of a 10 MW/4 h Carnot battery. Recent developments indicate a trend toward multi-dimensional integration. Frate et al. [25] proposed a dynamic co-simulation method based on machine characteristic curves, coupling compressor and turbine isentropic efficiency curves with an unsteady heat transfer model of the thermal energy storage unit, enabling dynamic simulation under varying operating conditions via real-time updating of working fluid properties. Nevertheless, existing approaches still face challenges: steady-state models often lack accuracy and adaptability, while dynamic models tend to focus on component-level behavior, neglecting system-level interactions.

Due to the bidirectional electro-thermal energy flow characteristics of Carnot batteries, a single efficiency indicator is insufficient for comprehensively evaluation. Consequently, multidimensional evaluation frameworks have been developed. Gonzalez et al. [26] developed a multi-parameter Pareto frontier model to optimize efficiency-power trade-offs, achieving a 49% improvement in round-trip efficiency compared with internally reversible models. Jiang et al. [27] proposed a multidimensional evaluation framework for TI-CB incorporating energy efficiency, exergy loss, economic performance, and energy storage density. Their exergy loss-based location analysis offers optimization insights for industrial waste heat recovery and grid energy storage applications. Chen et al. [28] conducted a multi-parameter coupling analysis focusing on key variables such as maximum temperature, impeller mechanical efficiency, and pressure ratio. Kurşun et al. [29] established a thermo-economic coupled model using round-trip efficiency and levelized cost of storage as core evaluation metrics, revealing synergistic optimization effects enabled by diversified heat sources. Although these studies successfully establish comprehensive static multidimensional evaluation frameworks that capture multiple performance dimensions, they generally rely on steady-state assumptions and fail to capture the dynamic nature of industrial operating conditions, where boundary parameters fluctuate continuously and systems frequently operate under off-design conditions.

In summary, Table 1 presents an overview of related studies. Existing research demonstrates that TI-CB systems offer superior potential for industrial waste heat recovery compared with alternative energy storage technologies. However, several critical research gaps remain. First, most studies rely on static evaluation frameworks that fail to capture the dynamic nature of industrial waste heat conditions, where boundary parameters fluctuate continuously due to process variations. Second, current modeling approaches lack comprehensive off-design analysis capabilities, despite TI-CB systems operating under non-nominal conditions for extended periods. Third, system design parameters often rely on empirical knowledge, with limited investigation into the influence of key variables on overall system performance. To address these gaps, this paper proposes a quasi-dynamic model that separately analyzes TI-CB operation under design and off-design conditions. It further develops a dynamic evaluation framework incorporating charging and discharging durations to assess overall system performance. Finally, it systematically investigates the impacts of key variables on TI-CB performance.

Based on these considerations, this study undertakes the following investigations:

• A quasi-dynamic model of the Carnot battery is developed, in which the system is divided into design and off-design operating conditions. This model integrates steady-state modeling with component characteristic maps, providing the dynamic adaptability required to simulate system performance under realistic industrial

Table 1 Overview of related works

Ref.	Steady-state modeling	Quasi-dynamic Modeling	Dynamic modeling	Parameter influence on performance	static evaluation framework	Dynamic evaluation framework	Summary
[15−17]	√	×	×	×	√	×	Modeling methods cannot reflect the performance changes when there are load fluctuations or when operating conditions deviate
[18−25]	×	×	√	×	√	×	The model is complex, with high computational cost, numerous parameters, and difficulty in solving
[26, 27]	√	×	×	×	√	×	The evaluation framework cannot fully reflect the changes in energy quality and the laws of performance evolution of Carnot battery during different phases
Wang et al. [28]	×	×	√	√	√	×	Although a systematic analysis has been conducted on some parameters, the impact of the main boundary parameters on the system is lacking
This paper	×	√	×	√	×	√	Proposed a quasi-dynamic model and a dynamic evaluation framework to systematically investigate parameter influence

operating conditions while retaining high computational efficiency.

• A dynamic evaluation framework that accounts for time-related effects is established to better characterize differences in charging and discharging durations and the performance variations induced by off-design operating conditions.

• Joint sampling and model validation techniques are employed to construct a multi-operating condition set. The characteristics of the Carnot battery under both design and off-design conditions are further analyzed to evaluate its adaptability and performance. Based on this multi-condition set, the effects of key boundary condition parameters, equipment performance parameters, and design parameters on Carnot battery performance are systematically investigated.

2 Methodology

2.1 System description

The TI-CB developed in this study for waste heat recovery in industrial integrated energy systems is illustrated in Fig. 1. R1233zd(E) is selected as the working fluid for both the heat pump (HP) and the ORC. The temperature range of the available industrial waste heat is 60–90 °C. The detailed *T*–*s* diagram and system operation processes are illustrated in Fig. 2. Table 2 summarizes the efficiencies of the major components and the key parameters used in the simulations, while Table 3 lists the equipment performance parameters and their corresponding values.

The TI-CB system consists of four primary heat exchangers, all modeled as a counter-flow heat exchanger. The HP cycle utilizes an axial compressor driven by an asynchronous motor, whereas the ORC power block employs a gas turbine or steam turbine coupled to an asynchronous generator. The ORC working fluid pump is modeled as a centrifugal pump.

During the energy charging phase, the thermodynamic evolution of the working fluid within HP is represented by a closed loop consisting of four state points. The HP cycle comprises four thermodynamic processes with corresponding state transitions. The refrigerant undergoes isobaric evaporation from subcooled liquid (State 1) to superheated vapor (State 2) during process 1–2. A non-isentropic compression process then elevates the vapor to

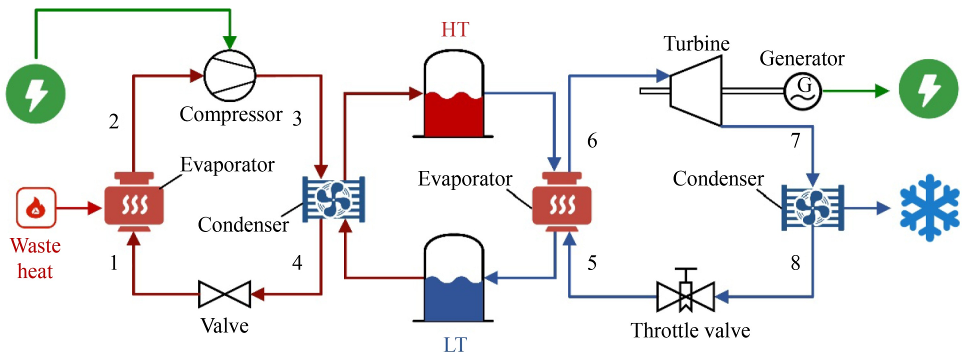

Fig. 1 System structure of TI-CB.

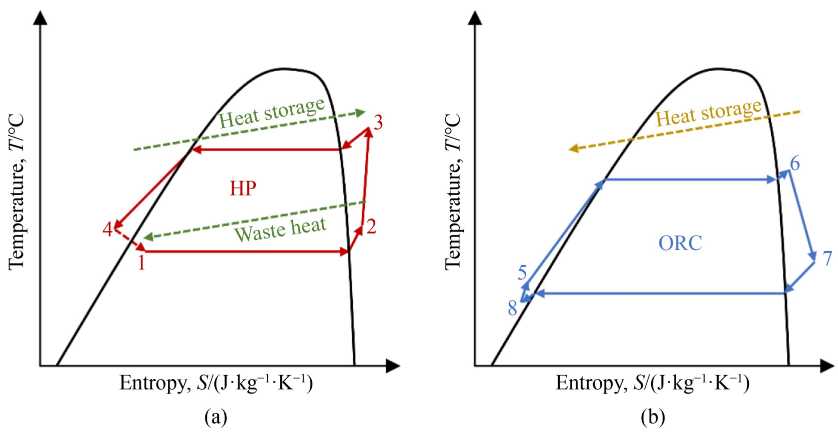

Fig. 2 *T*–*s* diagrams of (a) HP and (b) ORC in TI-CB.

Table 2 Design parameters of the TI-CB

Phase	Variables	Symbol
Energy charging	Evaporation pressure (temperature) of HP	$P^{hp}_{evap}(T^{hp}_{evap})$
	Overheat degree of evaporator in HP	ΔT^{hp}_{sup}
	Condensation pressure (temperature) of HP	$P^{hp}_{cond}(T^{hp}_{cond})$
	Overcool degree of condenser in HP	ΔT^{hp}_{sub}
	Mass flow in HP	m^{hp}
Thermal storage	High temperature tank thermal storage temperature	T^{sto}_{h}
	High temperature tank thermal storage pressure	P^{sto}_{h}
	Low temperature tank thermal storage temperature	T^{sto}_{c}
	Low temperature tank thermal storage pressure	P^{sto}_{c}
	Charging duration	τ_{ch}
	Discharging duration	τ_{dis}
Energy discharging	Evaporation pressure (temperature) of ORC	$P^{orc}_{evap}(T^{orc}_{evap})$
	Overheat degree of evaporator in ORC	ΔT^{orc}_{sup}
	Condensation pressure (temperature) of ORC	$P^{orc}_{cond}(T^{orc}_{cond})$
	Overcool degree of condenser in ORC	ΔT^{orc}_{sub}
	Mass flow in ORC	m^{orc}
Boundary conditions	Inlet temperature of the waste heat source	T^{hs}_{in}
	Outlet temperature of the waste heat source	T^{hs}_{out}
	Inlet pressure of the waste heat source	P^{hs}_{in}
	Inlet mass flow of the waste heat source	m^{hs}_{in}
	Inlet temperature of the ambient cold source	T^{cs}_{in}
	Outlet temperature of the ambient cold source	T^{cs}_{out}

Table 3 Equipment performance parameters of TI-CB

Variables	Symbol	Value
Isentropic efficiency of compressor	η^{comp}_{s}	0.8
Isentropic efficiency of turbine	η^{turb}_{s}	0.85
Isentropic efficiency of pump	η^{pump}_{s}	0.75
Efficiency of motor	η^{motor}	0.98
Efficiency of generator	η^{gen}	0.98
Pressure ratio of piping and heat exchangers	pr	0.98
Temperature pinch point in evaporator	ΔT^{evap}_{pp}	5 °C
Temperature pinch point in condenser	ΔT^{cond}_{pp}	5 °C

high-pressure, high-temperature conditions (State 3) in process 2–3. The refrigerant subsequently releases heat isobarically in the condenser and condenses to a subcooled liquid (State 4) during process 3–4. Finally, an isenthalpic throttling process completes the cycle in process 4–1, returning the working fluid to its initial state.

During the energy discharging phase, the thermodynamic evolution of the working fluid in ORC is described by a sequence of key state points. The ORC also operates through four sequential thermodynamic processes. The organic working fluid absorbs heat isobarically, transitioning from subcooled liquid (State 5) to superheated vapor (State 6) during process 5–6. Non-isentropic expansion in the turbine generates work output while reducing the pressure and temperature to State 7 through process 6–7. Isobaric condensation follows in process 7–8, returning the fluid to a subcooled liquid state (State 8), while isentropic pumping completes the cycle through process 8–5, pressurizing the liquid back to the initial state.

2.2 Quasi-dynamic model

In the quasi-dynamic modeling approach, the three phases of the TI-CB —energy charging, thermal storage, and energy discharging—are decoupled to significantly reduce the complexity of parameter identification in multi-energy flow coupled modeling, while enabling independent performance optimization of each functional module. Under design operating conditions, i.e., steady-state modeling, all processes are formulated based on the fundamental conservation laws of thermodynamics, including conservation of energy, mass, and momentum. By defining core system parameters, energy balance and heat flux continuity equations are established. The modeling framework is developed under the following assumptions:

• Energy charging: A quasi-static compressor characteristic model is coupled with steady-state heat transfer equations to represent the thermodynamic behavior of the charging process.

• Thermal storage: A thermal capacity-flow rate balance equation is constructed under a uniform temperature assumption, thereby simplifying model complexity by neglecting temperature stratification effects.

• Energy discharging: A quasi-steady description of heat-to-power conversion is achieved by interpolating the thermophysical properties of the working fluid across different phase states, under steady-flow assumptions.

Under off-design operating conditions, characteristic curve techniques are introduced to represent the nonlinear behavior of system components under variable operating regimes. This approach captures the dynamic performance variations—particularly in efficiency and pressure loss—of key subcomponents within the Carnot battery as influenced by fluctuations in the working fluid mass flow rate.

a Energy charging

(a) Heat exchangers

The steady-state models of the evaporator and condenser are constructed based on the first law of thermodynamics in conjunction with heat transfer equations

$$h_{\mathrm{steps}} = [h_{\mathrm{in}}, h_{\mathrm{sat}_{\mathrm{gas}}}, h_{\mathrm{sat}_{\mathrm{liquid}}}, h_{\mathrm{out}}], \tag{1}$$

where h_{steps} denotes the enthalpy node values for segmentation (J/kg), $h_{\mathrm{sat}_{\mathrm{gas}}}$ is defined as the enthalpy of saturated gas phase (J/kg), $h_{\mathrm{sat}_{\mathrm{liquid}}}$ is defined as the enthalpy of saturated liquid phase (J/kg), h_{in} denotes the inlet enthalpy (J/kg), and h_{out} stands for the outlet enthalpy (J/kg).

The heat transfer rate of the heat exchanger can be calculated by the following equation

$$\dot{Q} = \mathrm{UA} \cdot \mathrm{LMTD}, \tag{2}$$

where $\dot{Q}$ denotes the heat transfer rate of the heat exchanger, UA represent the overall heat transfer coefficient (K), and LMTD is the logarithmic mean temperature difference (K), which is typically calculated as

$$\ddot{A}T_{\log,i} = \frac{(T_{\mathrm{hot},i} - T_{\mathrm{cold},i+1}) - (T_{\mathrm{hot},i+1} - T_{\mathrm{cold},i})}{\ln\left(\dfrac{T_{\mathrm{hot},i} - T_{\mathrm{cold},i+1}}{T_{\mathrm{hot},i+1} - T_{\mathrm{cold},i}}\right)}, \tag{3}$$

where $T_{\mathrm{hot},i}$ is the hot-side fluid temperature of the heat exchanger (K), and $T_{\mathrm{cold},i}$ is the cold-side fluid temperature (K). For off-design condition, UA can be expressed as

$$\mathrm{UA}_{\mathrm{off}} = \mathrm{UA} \cdot f_{\mathrm{char}}\left(\frac{\dot{m}}{m}\right), \tag{4}$$

where f_{char} denotes the characteristic map, $\dot{m}$ represents the mass flow rate under off-design conditions. The pressure ratio across the heat exchanger is also influenced by variations in the working fluid mass flow rate, which reflects the hydraulic performance of the heat exchanger and can be expressed as [30]

$$pr = \sqrt{1 - \frac{\zeta \cdot \dot{m}^2}{2 \cdot \rho \cdot A^2 \cdot p_{\mathrm{in}}}}, \tag{5}$$

where ζ denotes the dimensionless flow resistance coefficient independent of geometry (m^{-4}), A represents the cross-sectional area of the flow channel (m^2).

(b) Compressor

Under design conditions, the isentropic efficiency of the compressor is expressed as

$$\eta_{\mathrm{s}} = \frac{h_{\mathrm{out,s}} - h_{\mathrm{in}}}{h_{\mathrm{out}} - h_{\mathrm{in}}}, \tag{6}$$

where η_{s} denotes the isentropic efficiency.

When operating under off-design conditions, a two-dimensional performance map correction approach is employed. Two dimensionless parameters—reduced rotational speed and reduced mass flow rate—are introduced and defined as [31]:

$$X = \sqrt{\frac{T_{\mathrm{in,design}}}{T_{\mathrm{in}}}}, \tag{7}$$

$$Y = \frac{\dot{m}_{\mathrm{in}} \cdot p_{\mathrm{in,design}}}{\dot{m}_{\mathrm{in,design}} \cdot p_{\mathrm{in}} \cdot X}, \tag{8}$$

where X is the reduced rotational speed; $T_{\mathrm{in,\,design}}$ denotes the working fluid inlet temperature at design conditions (K), T_{in} is the inlet temperature under actual operating conditions (K); Y is the reduced mass flow rate, $\dot{m}_{\mathrm{in,design}}$ and $\dot{m}_{\mathrm{in}}$ are the working fluid inlet mass flow rates at design and actual conditions, respectively (kg/s); and $p_{\mathrm{in,\,design}}$ and p_{in} are the inlet pressures at design and actual conditions (Pa). Based on these definitions, the isentropic efficiency under off-design conditions can be computed as

$$\eta_{\mathrm{s}} = \eta_{\mathrm{s,design}} \cdot g(X, Y) \cdot \left(1 - \frac{\mathrm{igva}^2}{10000}\right), \tag{9}$$

where igva is the inlet guide vane angle, which is taken as 0 in this paper, so that the compressor's off-design performance is determined solely by the characteristic map functions; $g(X, Y)$ represents the compressor efficiency performance mapping function.

Similarly, the pressure ratio under off-design conditions can be obtained by

$$pr = pr_{\mathrm{design}} \cdot f\,(X, Y) \cdot \left(1 - \frac{\mathrm{igva}}{100}\right), \tag{10}$$

where $f\,(X, Y)$ is the characteristic pressure ratio mapping function.

(c) Throttle valve

Under design conditions, the core assumption for the throttle valve is that no energy exchange occurs during the throttling process. The corresponding expression is given as

$$h_{\mathrm{out}} = h_{\mathrm{in}}. \tag{11}$$

Pressure drop and pressure ratio are expressed as

$$pr = \frac{p_{\mathrm{out}}}{p_{\mathrm{in}}}, \tag{12}$$

$$\text{Ä}p = p_{\mathrm{in}} \cdot (1 - pr), \tag{13}$$

where pr is the pressure ratio, p_{out} denotes the outlet pressure of the fluid (Pa), p_{in} denotes the inlet pressure of the fluid (Pa), and $\text{Ä}p$ represents the pressure drop (Pa).

Under off-design operating conditions, the actual pressure drop is influenced by fluctuations in the working fluid mass flow rate, as described by

$$\text{Ä}p = p_{\mathrm{in}} - p_{\mathrm{out}} = f\,(\dot{m}), \tag{14}$$

(d) Motor

To simplify the model, the dynamic response characteristics of the motor under grid frequency regulation conditions are considered. The energy conversion efficiency of the motor is defined as

$$\eta_{\mathrm{motor}} = \frac{P_{\mathrm{shaft}}}{P_{\mathrm{elec}}}, \tag{15}$$

where η_{motor} is the motor efficiency, P_{shaft} is the mechanical power output at the motor shaft (W), and P_{elec} denotes the input electrical power (W).

b Thermal storage

(a) Storage tank

The dual-tank thermal storage system serves as the thermal energy carrier of the Carnot battery. Its volume–pressure coordinated design ensures the thermodynamic stability of the storage medium during charge–discharge cycles. During the energy charging and discharging phases, the heat exchanged by the thermal storage medium is related to its mass flow rate and temperature gradient, and can be expressed as

$$\dot{Q}_{\mathrm{ch}} = \dot{m}_{\mathrm{ch}} c_{\mathrm{p}} \left(T_{\mathrm{hot,tank}} - T_{\mathrm{cold,tank}}\right), \tag{16}$$

$$\dot{Q}_{\mathrm{dis}} = \dot{m}_{\mathrm{dis}} c_{\mathrm{p}} \left(T_{\mathrm{hot,tank}} - T_{\mathrm{cold,tank}}\right), \tag{17}$$

In the equations, $\dot{Q}_{\mathrm{ch}}$, $\dot{Q}_{\mathrm{dis}}$ represent the heat stored and heat released by the Carnot battery under charging and discharging conditions (W), $\dot{m}_{\mathrm{ch}}$ and $\dot{m}_{\mathrm{dis}}$ denote the corresponding mass flow rates of the thermal storage fluid during charging and discharging (kg/s), and $T_{\mathrm{hot,\ tank}}$ and $T_{\mathrm{cold,\ tank}}$ are the thermal storage temperatures of the fluid in the hot and cold tanks (K).

(b) Pump

Under design conditions, similar to the compressor, the energy conversion efficiency of the pump is defined using the isentropic efficiency, as

$$\eta_{\mathrm{p}} = \frac{h_{\mathrm{out,s}} - h_{\mathrm{in}}}{h_{\mathrm{out}} - h_{\mathrm{in}}}, \tag{18}$$

where η_{p} is the isentropic efficiency of the pump. The shaft power of the pump is calculated as

$$P = \frac{\dot{m} \cdot (h_{\mathrm{out}} - h_{\mathrm{in}})}{\eta_{\mathrm{s}}}. \tag{19}$$

When operating under off-design conditions, the efficiency of the working fluid pump can be corrected using an empirical efficiency–flow rate characteristic correlation, as given by

$$\eta_{\mathrm{s}} = \eta_{\mathrm{s,design}} \cdot g\left(\frac{\dot{V}}{\dot{V}_{\mathrm{design}}}\right), \tag{20}$$

where $g\,(\cdot)$ denotes the characteristic mapping function of the pump's isentropic efficiency.

c Energy discharging

(a) Turbine

Under design operating conditions, the isentropic efficiency of the expander is defined as the ratio of the actual specific work output to that of an ideal isentropic expansion process, as given by

$$\eta_{\mathrm{t}} = \frac{h_{\mathrm{out}} - h_{\mathrm{in}}}{h_{\mathrm{out,s}} - h_{\mathrm{in}}}, \tag{21}$$

where η_{t} is the isentropic efficiency of the turbine.

Similar to the compressor model, the isentropic efficiency of the expander under off-design conditions can be calculated as [31]

$$\eta_{\mathrm{s}} = \eta_{\mathrm{s,design}} \cdot f\left(\frac{\dot{m}}{\dot{m}_{\mathrm{design}}}\right). \tag{22}$$

Stodola's cone law is used to describe the flow characteristics of a turbine under off-design conditions [32]:

$$\dot{m}_{\mathrm{in}} = \frac{\dot{m}_{\mathrm{in,design}} \cdot p_{\mathrm{in}}}{p_{\mathrm{in,design}}} \cdot \sqrt{\frac{p_{\mathrm{in,design}} \cdot v_{\mathrm{in}}}{p_{\mathrm{in}} \cdot v_{\mathrm{in,design}}} \cdot \frac{1 - \left(\dfrac{p_{\mathrm{out}}}{p_{\mathrm{in}}}\right)^2}{1 - \left(\dfrac{p_{\mathrm{out,design}}}{p_{\mathrm{in,design}}}\right)^2}}, \tag{23}$$

where the subscript design denotes the value at the design condition, and v is the specific volume.

(b) Generator

Similar to the motor model, the dynamic response characteristics of the generator during grid frequency regulation—such as transient behavior and control dynamics—are neglected to simplify the system model. The energy conversion efficiency of the generator is defined as

$$\eta_{\text{gen}} = \frac{P_{\text{elec}}}{P_{\text{shaft}}}, \tag{24}$$

where η_{gen} is the generator efficiency, P_{shaft} is the electrical power output (W), and P_{elec} is the mechanical shaft power input (W).

2.3 Thermodynamic solution method

The thermodynamic solution method employed in this paper is divided into two stages: the design stage and the off-design stage. At the design point, operating conditions are solved by constructing a comprehensive thermodynamic system model with predefined boundary conditions and component properties. The thermodynamically coupled components form a system of nonlinear equations, which is solved using the Newton–Raphson algorithm with Jacobian matrix updates until the convergence criteria are satisfied.

During the off-design solution stage, the geometric dimensions and inherent characteristics of key components are assumed to be fixed. Component performance maps (characteristic curves) are introduced, and interpolation techniques are used to capture their nonlinear behavior. These maps provide the necessary data support for solving the thermodynamic network under off-design conditions. When the operating points of TI-CB deviate from the design point, parameters such as pressure drop and isentropic efficiency—originally treated as fixed in the design-point solution—are no longer assumed constant. Instead, characteristic curves are utilized to dynamically update these parameters based on changes in the working fluid mass flow rate. The Newton–Raphson method is applied iteratively to solve the resulting nonlinear system.

This approach enables accurate determination of actual component performance and the overall thermodynamic cycle parameters of TI-CB under off-design operating conditions. The specific steps of the solution process are illustrated in Fig. 3.

2.4 Joint sampling method

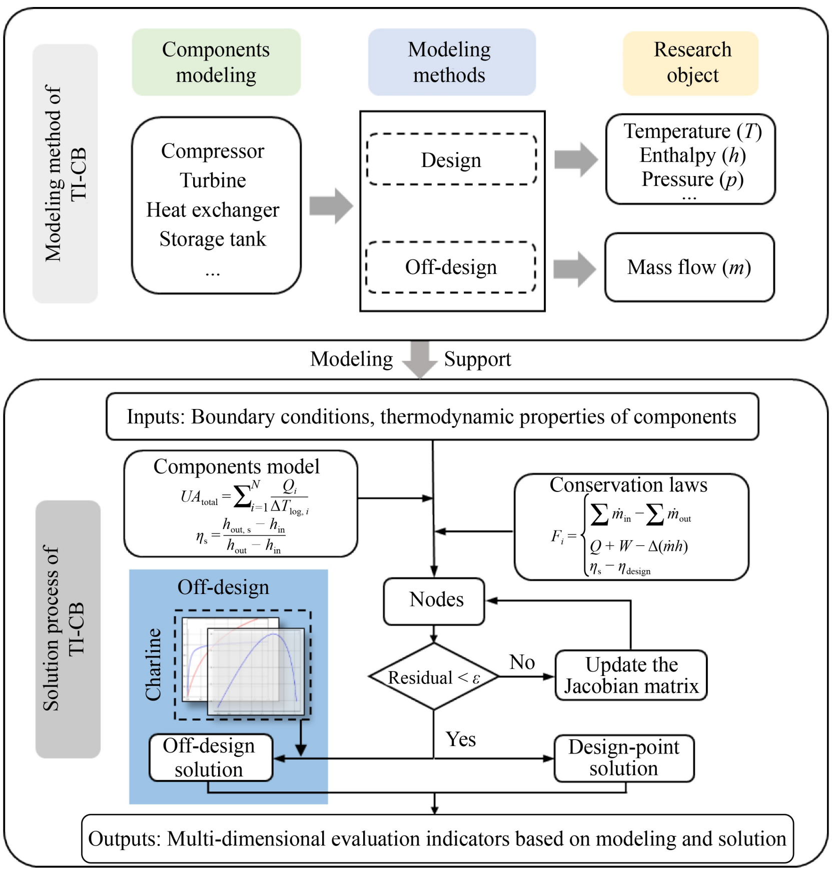

Fig. 3 Modeling and solution methodology for the TI-CB system.

The first step of the joint sampling method is to sample input variables. Input variables are sampled using a uniform distribution generated from a periodic curve. Each input variable is assigned a unique integer frequency and is sampled uniformly over a specified interval, as expressed in

$$u_j(s) = \frac{1}{2} + \frac{1}{\pi}\arcsin\left(\sin\left(\omega_j s + \varphi_j\right)\right), j \in [1,n], \tag{25}$$

where ω_j is the integer frequency associated with the input variable x_j, and φ_j is a randomly selected phase-shift parameter. To transform the uniformly distributed samples into samples that follow the desired input distribution, the inverse transform method is used, expressed as

$$x_j = F_{x_j}^{-1}(u_j), j \in [1,n], \tag{26}$$

where $F_{x_j}^{-1}$ is the inverse of the cumulative distribution function for the input variable x_j. The parameter space formed by the input variables is

$$\Theta = \{X_i | X_i = [x_{i,1}, x_{i,2}, \ldots, x_{i,j}, \ldots, x_{i,n}], i \in [1,n], j \in [1,n]\}, \tag{27}$$

where Θ denotes the parameter space, and X_i represents the parameter vector composed of the variable combinations. The next step is parameter sample validation, where physical constraints are imposed on the parameter vectors in the parameter space.

$$\begin{cases} g(T^{\mathrm{evap}}, T^{\mathrm{cond}}) = T^{\mathrm{evap}} - T^{\mathrm{cond}} \in [\Delta T_{\min}, \Delta T_{\max}] \\ \cdots \\ g(X_i) \in \omega_{\mathrm{phys}} \end{cases}, \tag{28}$$

where $g(\cdot)$ represents the temperature-zone-related constraints that the Carnot battery parameters must satisfy, defining the thermodynamically feasible domain. The feasibility of the samples in the parameter space is validated against the constraints

$$\Theta_{\mathrm{valid}} = \{X_i | f(X_i) \text{ converges and } g(X_i) \in \omega_{\mathrm{phys}}, i \in [1,n]\}, \tag{29}$$

where $f(\cdot)$ is the Carnot battery solver model. After validation, sampling interval reduction is conducted. For the parameters validated in the parameter space, a new sample space Θ_{valid} is constructed

$$\begin{aligned} \Gamma = \{X'_j | X'_j = [x_{1,j}, x_{2,j}, \ldots, x_{i,j}, \ldots, x_{m,j}] \\ \text{and } X_i \in \theta_{\mathrm{valid}}, i \in [1,m], j \in [1,n]. \end{aligned} \tag{30}$$

The probability distribution function of the variables in the sample space Γ can be expressed as

$$P(X'_j = x_{m,j}) = p_m, \tag{31}$$

Additionally, the probabilities satisfy the condition $\sum_m p_m = 1$. When a probability density function threshold ϵ is set, parameter samples are discarded if $p_m < \epsilon$. After this sample compression, the upper and lower bounds of the parameter sampling interval are reset. Resampling is then performed following the initial sampling procedure to generate a new set of parameter space samples that better conform to the thermodynamic operating constraints of the Carnot battery. This refined sample set is subsequently used to construct the multi-operating condition set for the Carnot battery.

2.5 Dynamic evaluation indicators

Building upon the TI-CB model and solution method developed in Section 2, this section establishes a dynamic evaluation framework for assessing the system performance. The proposed dynamic evaluation indicators, based on time integration, capture the time-varying energy and exergy flows during charging and discharging, enabling a comprehensive performance assessment under multi-load and unsteady-state conditions. Unlike static indicators, these dynamic indicators account for both the quantity and quality of energy, more accurately reflecting actual operational characteristics.

Based on the energy and exergy conversion pathways of the Carnot battery—electrical to thermal and thermal to electrical—general energy valuation indicators for the Carnot battery are

$$\eta_{\mathrm{E2H}} = \frac{Q_{\mathrm{in}}}{E_{\mathrm{in}}}, \tag{32}$$

$$\eta_{\mathrm{sto}} = \frac{Q_{\mathrm{out}}}{Q_{\mathrm{in}}} = 1 - \frac{\ddot{\mathrm{A}}Q}{Q_{\mathrm{in}}}, \tag{33}$$

$$\eta_{\mathrm{H2E}} = \frac{E_{\mathrm{out}}}{Q_{\mathrm{out}}}, \tag{34}$$

$$\eta_{\mathrm{rt}} = \frac{E_{\mathrm{out}}}{E_{\mathrm{in}}} = \frac{Q_{\mathrm{in}}}{E_{\mathrm{in}}} \cdot \frac{Q_{\mathrm{out}}}{Q_{\mathrm{in}}} \cdot \frac{E_{\mathrm{out}}}{Q_{\mathrm{out}}} = \eta_{\mathrm{E2H}} \cdot \eta_{\mathrm{sto}} \cdot \eta_{\mathrm{H2E}}, \tag{35}$$

$$\eta_{\mathrm{ex}} = \frac{\dot{E}_{\mathrm{out}} + \mathit{\Delta}\dot{E}_{\mathrm{sys}}}{\dot{E}_{\mathrm{in}}}, \tag{36}$$

where η_{E2H} represents the electrical-to-thermal efficiency in the energy charging phase of the Carnot battery; η_{sto} denotes the thermal storage efficiency in the thermal storage phase; η_{H2E} is the thermal-to-electrical efficiency in the energy discharging phase; η_{rt} is the round-trip efficiency of the Carnot battery, E_{in} denotes the electrical energy consumed in the thermal storage phase; Q_{in} represents the thermal energy transferred to the thermal storage system in the energy charging phase; $\ddot{\mathrm{A}}Q$ is the heat loss in the thermal storage phase; Q_{out} denotes the thermal energy absorbed from the thermal storage phase during the energy discharging phase; E_{out} denotes the electrical energy generated in the energy discharging phase; $\dot{E}_{\rm out}$ represents the output exergy of the Carnot battery during the energy discharging phase, which may take the form of power supplied externally by ORC or heat provided through a heat exchanger (J); $\Delta\dot{E}_{\rm sys}$ denotes the thermal exergy stored in the Carnot battery's thermal storage system, (J); and $\dot{E}_{\rm in}$ represents the input exergy of the Carnot battery during the energy charging phase, primarily consisting of consumed electrical energy, but may also include the thermal exergy of waste heat resources, (J).

Static round-trip efficiency and exergy efficiency indicators often fail to accurately characterize Carnot battery performance in industrial integrated energy systems under varying operating conditions. To address this, dynamic performance indicators are proposed, which account for the time differences in energy charging and discharging, as well as performance variations induced by off-design conditions

$$\eta_{\rm rt}=\frac{\int_0^{\tau_{\rm dis}} m_{\rm orc}(t)\cdot\left(w^{\rm turb}(t)\cdot\eta^{\rm gen}(t)-\dfrac{w^{\rm pump}(t)}{\eta^{\rm motor}(t)}\right)\mathrm{d}t}{\int_0^{\tau_{\rm ch}} m_{\rm hp}(t)\cdot\dfrac{w^{\rm comp}(t)}{\eta^{\rm motor}(t)}\mathrm{d}t}, \tag{37}$$

$$\eta_{\rm ex}=\frac{\int_0^{\tau_{\rm dis}} m_{\rm orc}(t)\cdot\left(w^{\rm turb}(t)\cdot\eta^{\rm gen}(t)-\dfrac{w^{\rm pump}(t)}{\eta^{\rm motor}(t)}\right)\mathrm{d}t}{\int_0^{\tau_{\rm ch}}\left(m_{\rm hp}(t)\cdot\dfrac{w^{\rm comp}(t)}{\eta^{\rm motor}(t)}+m_{\rm hs}(t)\cdot\left(e_{\rm in}^{\rm hs}(t)-e_{\rm out}^{\rm hs}(t)\right)\right)\mathrm{d}t}, \tag{38}$$

where $\tau_{\rm ch}$ represents the operating time of the Carnot battery under energy charging (s); $\tau_{\rm dis}$ denotes the operating time of the Carnot battery during energy discharging (s); $m_{\rm hp}(t)$ is the mass flow rate of the working fluid in HP at time t, (kg/s); $m_{\rm orc}(t)$ is the mass flow rate of the working fluid in the ORC cycle at time t, (kg/s); $W^{\rm comp}(t)$ is the specific power of the compressor in the HP at time t, (W/kg); $w^{\rm turb}$ is the specific power of the expander in the ORC at time t, (W/kg); $w^{\rm pump}$ is the specific power of the working fluid pump in the ORC at time t, (W); $\eta^{\rm gen}(t)$ is the efficiency of the generator at time t; $\eta^{\rm motor}(t)$ is the efficiency of the motor at time t; $m_{\rm hs}(t)$ represents the mass flow rate of the waste heat resource at time t, (kg/s); $e_{\rm in}^{\rm hs}$ denotes the specific exergy of the waste heat resource at the inlet, (W/kg); and $e_{\rm out}^{\rm hs}$ denotes the specific exergy of the waste heat resource at the outlet, (W/kg).

3 Model validation

The case study models in this paper are developed using Python on a Windows 11 operating system, with an Intel® Core™ i7-12700H processor and 16 GB of RAM. The Carnot battery system modeling and solution are performed using the TesPy library for network convergence calculations, while a custom fluid solver engine is implemented by dynamically linking to REFPROP 10.0 to query and solve for the thermophysical properties of the working fluid.

To validate the effectiveness of the modeling approach, a comparison is made with Cooke et al. [33], which also investigates the waste heat recovery performance of TI-CB. In both studies, the heat source and cold source are water, and the thermal storage medium is water at atmospheric pressure. The working fluids for the HP and ORC are both R1233zd(E). Identical model assumptions and boundary conditions are applied. The simulation results provide temperature, pressure, and enthalpy values at various operating points of the Carnot battery, as shown in Table 4. The calculated performance indicators are compared with those from the reference literature, as presented in Table 5. The minimal errors

Table 4 Comparison of operating point parameters of the Carnot battery

Points	Temperature/°C		Pressure/kPa		Enthalpy/(kJ·kg^{-1})	
	Hu et al. [33]	Simulation	Hu et al. [33]	Simulation	Hu et al. [33]	Simulation
1	50.7	50.7	299.4	299.0	308.9	302.2
2	53.9	52.7	299.4	299.0	441.6	440.5
3	106.8	105.1	1154	1162.0	474.3	471.8
4	86.7	81.8	1154	1154.0	308.9	302.2
5, 8	81.7	81.7	106.4	106.4	342.4	342.4
6,7	100	100.0	106.4	106.4	419.2	419.4
9	83.9	84.9	705.2	704.6	460.5	461.5
10	43.1	44.2	153.2	153.1	435.7	436.6
11	29.7	27.8	153.2	153.1	235.7	233.3
12	30	28.1	705.2	704.6	236.3	233.9

observed demonstrate that the accuracy of the modeling and solution methods meets the engineering requirements, confirming the reliability of the approach.

Additionally, to further validate the robustness of the model, a second comparative validation is conducted against Jiang et al. [27], which investigates an enhanced TI-CB by installing with installed recuperators. A comparison is made with the basic TI-CB, which has the same configuration as in this paper, and the results are presented in Table 6.

4 Results and discussion

4.1 Multi-operating condition set

This study focuses on recovering waste heat from industrial wastewater within the 60–90°C temperature range using Carnot battery in integrated energy systems. A multi-operating-condition set for the TI-CB is constructed by combining design parameters to establish nominal operating conditions with perturbation parameters to account for external influences under off-design conditions. Because the Carnot battery involves fluids with varying energy grades during the energy charging, thermal storage, and energy discharging phases, a variable transformation approach is employed to integrate operational constraints into design variables. The initial upper and lower bounds for the transformed design parameters are established, as detailed in Table 7.

Uniform sampling was performed over 8 design variables, generating a sample space of 8000 points. After feasibility screening, the sample space was reduced to 5110 feasible samples. The design parameters were normalized, and their distributions are illustrated in Fig. 4. Among these, ΔT_3^{hp}, ΔT_3^{orc}, ΔT_2^{orc} exhibit good uniformity, primarily because they are independent input variables. In contrast, significant distortion in the sample uniformity is observed for $\Delta T^{hs}, \Delta T_2^{hp}$ and ΔT^{sto}. This distortion arises from the coupled nature of the system's thermal interactions. Specifically, ΔT^{hs} and ΔT_2^{hp} represent the temperature drop of the waste heat resource and the temperature difference between the condenser and evaporator in the HP, respectively. These parameters directly affect the condensation temperature of the HP, and subsequently the temperature in the HT. In the ORC, since the ambient cooling source is fixed and the temperature drop from condenser to evaporator is constrained to maintain η_{orc}, the LT has a lower bound. Consequently, when ΔT^{hs} approaches its upper sampling limit and ΔT_2^{hp} approaches its lower limit, the HT temperature becomes very low. In such cases, if ΔT^{sto} simultaneously approaches its upper sampling bound, the corresponding samples are likely to violate system constraints, leading to noticeable distortion in the feasible sample distribution.

Based on the initial sample distribution, the bounds of key design variables were adjusted: the upper limit of

Table 5 Comparison of Carnot battery performance with Hu et al. [33]

Parameters	Hu et al. [33]	Simulation
Mass flow of HP/(kg·s^{-1})	46.3	45.5
Mass flow of ORC/(kg·s^{-1})	32.4	32.4
Compressor power consumption	1545	1453
Expander power output	789	790
COP	4.96	5.07
η_{orc}	10%	10.4%
η_{rt}	50%	53.5%

Table 6 Comparison of Carnot battery performance with Jiang et al. [27]

Parameters	Jiang et al. [27]	Simulation
COP	5.164	5.09
η_{orc}	13.3%	11%
η_{rt}	61.71%	59.1%

Table 7 Design variable transformation relationships

Phase	Symbol	Transformation relation	Bounds/°C
Energy charging	T_{in}^{hs}	–	[60–90]
	ΔT^{hs}	$\Delta T^{hs} = T_{in}^{hs} - T_{out}^{hs}$	[10–40]
	ΔT_1^{hp}	$\Delta T_1^{hp} = T_{out}^{hs} - T_{evap}^{hp}$	[5–10]
	ΔT_2^{hp}	$\Delta T_2^{hp} = T_{cond}^{hp} - T_{evap}^{hp}$	[20–60]
Thermal storage	ΔT_3^{hp}	$\Delta T_3^{hp} = T_{cond}^{hp} - T_h^{sto}$	[5–10]
	ΔT^{sto}	$\Delta T^{sto} = T_h^{sto} - T_c^{sto}$	[10–40]
Energy discharging	ΔT_1^{orc}	$\Delta T_1^{orc} = T_c^{sto} - T_{evap}^{orc}$	[5–10]
	ΔT_2^{orc}	$\Delta T_2^{orc} = T_{cond}^{orc} - T_{out}^{cs}$	[5–10]

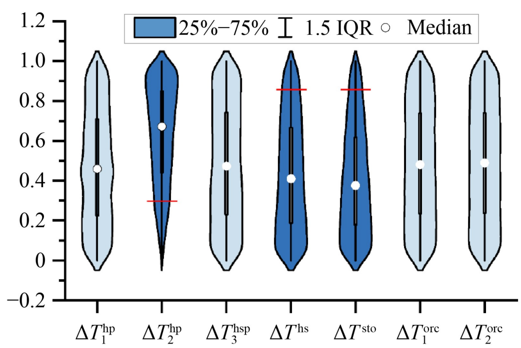

Fig. 4 Normalized parameter distributions in the feasible sample space from initial sampling.

ΔT^{hs} was set to 25 °C, the lower limit of ΔT_2^{hp} to 40 °C, and the upper limit of ΔT^{sto} was also revised. A new uniform sampling generated 8000 samples, of which 7605 remained after feasibility filtering, indicating good validity. The initial upper and lower bounds for the transformed design parameters are detailed in Table 5. These design variables effectively serve as the thermodynamic constraints between components, ensuring that all 7605 feasible samples analyzed in this study are thermodynamically consistent. After normalization, the distributions of the design parameters shown in Fig. 5 demonstrate improved uniformity, supporting the reliability of the subsequent sensitivity analysis.

4.2 Sensitivity analysis

The resampled design parameter sets were input into the Carnot battery model to compute the corresponding performance metrics: COP, η_{orc}, η_{rt} and η_{ex}. Fourier amplitude sensitivity testing (FAST) was applied to obtain first-order and total-order sensitivity indices under a 95% confidence interval. The results of the sensitivity analysis are shown in Fig. 6.

For COP, which characterizes the energy charging phase, the most influential parameter is ΔT_2^{hp}—the

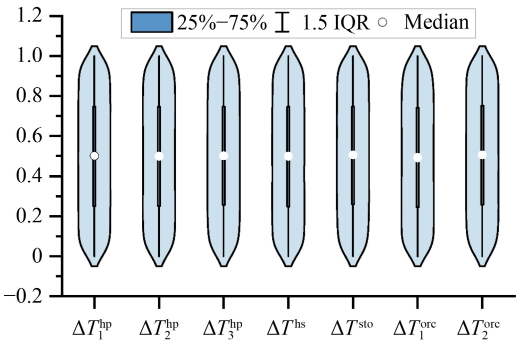

Fig. 5 Normalized parameter distributions in the feasible sample space from resampling.

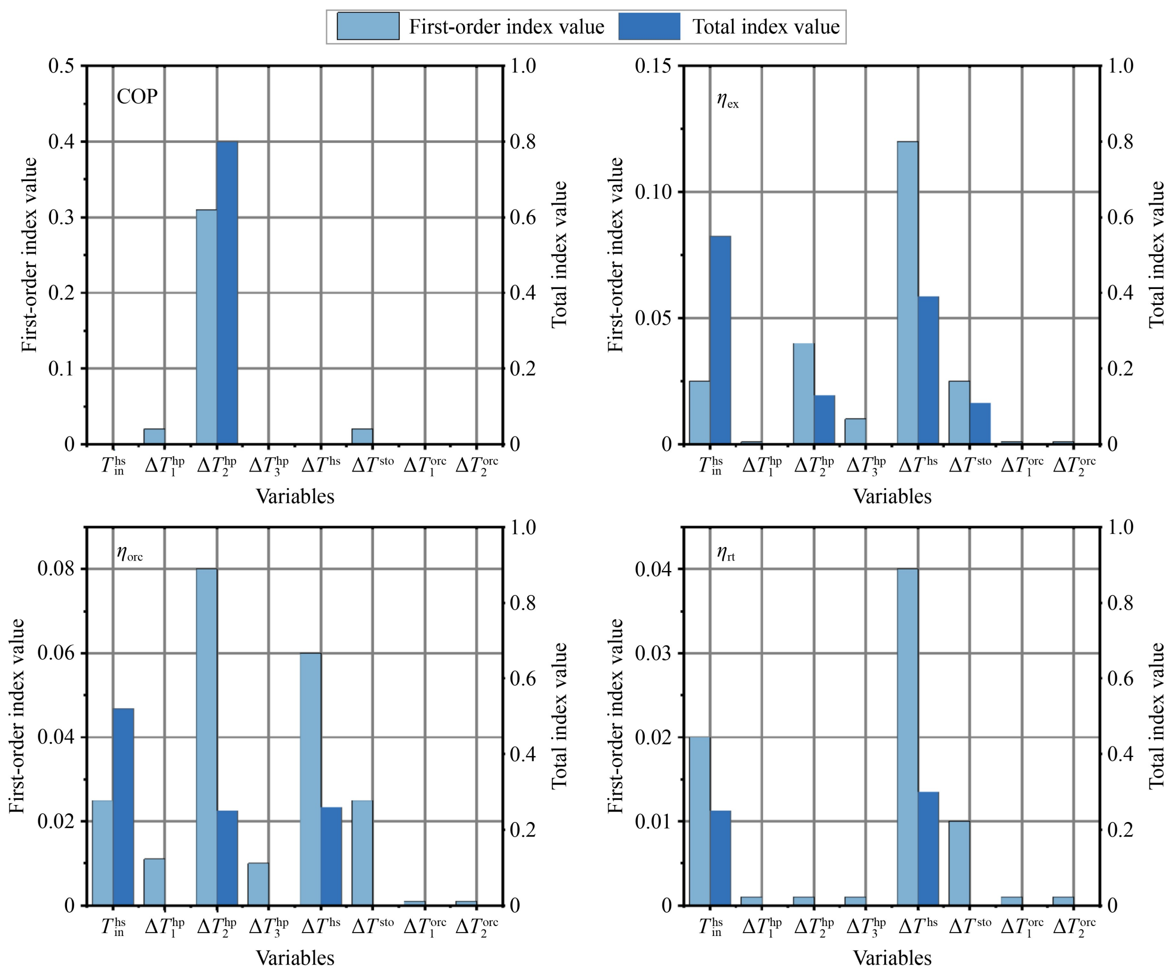

Fig. 6 Sensitivity analysis of design variables on performance indicators.

temperature difference between the condenser and evaporator. This is because the condensation temperature is closely related to the heat exchange process in the evaporator, affecting the amount of heat delivered outward, while the temperature difference also reflects the pressure ratio, which directly impacts compressor power consumption. For η_{orc} in the energy discharging phase of TI-CB, the most influential parameters are $T_{\mathrm{in}}^{\mathrm{hs}}$, ΔT^{hs}, and ΔT_2^{hp}. Specifically, $T_{\mathrm{in}}^{\mathrm{hs}}$ and ΔT^{hs} jointly determine the amount of heat absorbed from the waste heat source by the HP, while ΔT_2^{hp} governs the COP of the HP. Together, these three parameters influence the thermal energy delivered to the storage system and the resulting high-temperature tank temperature. For η_{ex}, the most influential parameters are $T_{\mathrm{in}}^{\mathrm{hs}}$, ΔT^{hs}, and ΔT^{sto}. These variables directly affect the exergy supplied to both the heat pump evaporator and ORC evaporator, thereby determining the system's overall exergy efficiency. The sensitivity analysis of η_{rt} shows that ΔT^{hs} is the dominant factor, with the highest first-order and total sensitivity indices. $T_{\mathrm{in}}^{\mathrm{hs}}$ and ΔT^{sto} also have moderate influence, while other variables have negligible impact, indicating that heat source utilization and thermal storage range are key to improving system efficiency.

In summary, the key design parameters affecting TI-CB performance are primarily those related to waste heat utilization—$T_{\mathrm{in}}^{\mathrm{hs}}$ and ΔT^{hs}—as well as ΔT_2^{hp}, which is associated with the energy charging phase. It is worth noting that, since the ambient cold source temperature is fixed and the high-temperature tank temperature is constrained by the thermodynamic process of the HP during the energy charging phase, the parameter ΔT^{sto}, associated with the thermal storage phase, is strongly restricted within the validated design space. Consequently, its variation range is limited, and its influence on Carnot battery performance is less significant.

4.3 Characteristic analysis of design condition parameters

Based on the validated parameter sample space, the corresponding performance of the TI-CB is computed. Using the obtained multi-condition set, the impact of key design parameters on the TI-CB's performance under design conditions is explored. The approach involves dividing the parameter sampling range into several subintervals, performing aggregated statistical analysis of performance within each subinterval, and examining trends in performance variations. The influence of parameters is visually represented using box plots, where the box ranges from the lower to the upper quartile. Fluctuations in the trend are attributed to the uneven distribution of parameter samples across different subintervals.

The influence of the heat source temperature on the main performance indicators of the TI-CB is first examined, as shown in Figs. 7–9. As the heat source temperature increases, the mean values of η_{orc} rises from 6.8% to 10.1%, η_{rt} from 34.3% to 67.9%, and η_{ex} from 25.3% to 35.4%. Although the thermophysical properties of the working fluid vary nonlinearly with increasing heat source temperature, causing local fluctuations in the curves, the overall increase in heat source temperature provides higher-quality thermal energy. Consequently, under similar distributions of other parameters, the performance of the TI-CB shows an overall increasing trend.

Subsequently, the effect of heat source temperature difference (ΔT^{hs}) on system performance is analyzed. The impact of the heat source temperature difference on the main performance indicators of TI-CB is shown in Figs. 10–12. As the heat source temperature difference increases, the mean value of η_{rt} decreases from 62.6% to 45.8%, η_{orc} from 10.4% to 7.5%, and η_{ex} from 36.5% to

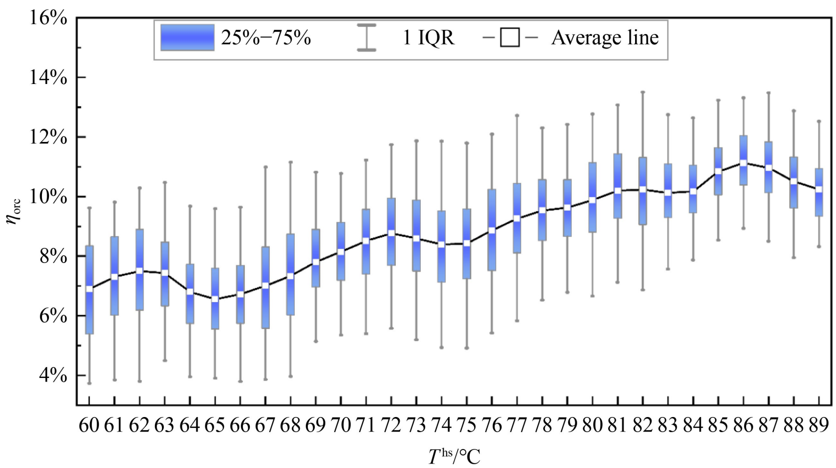

Fig. 7 Impact of heat source temperature on η_{orc}.

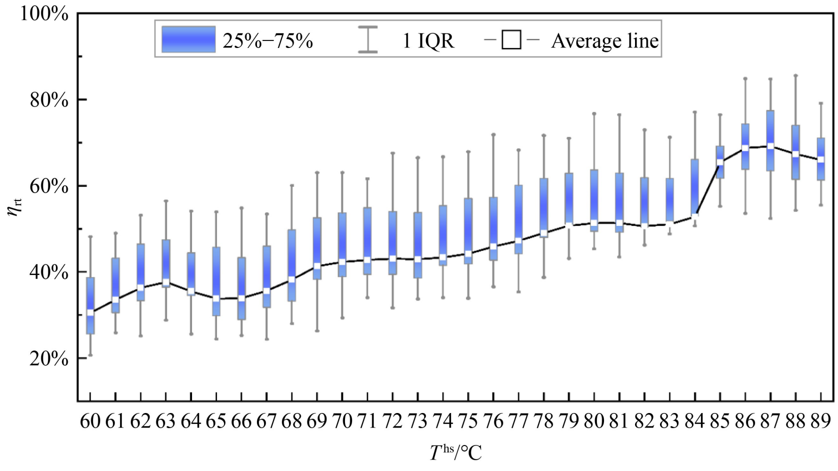

Fig. 8 Impact of heat source temperature on η_{rt}.

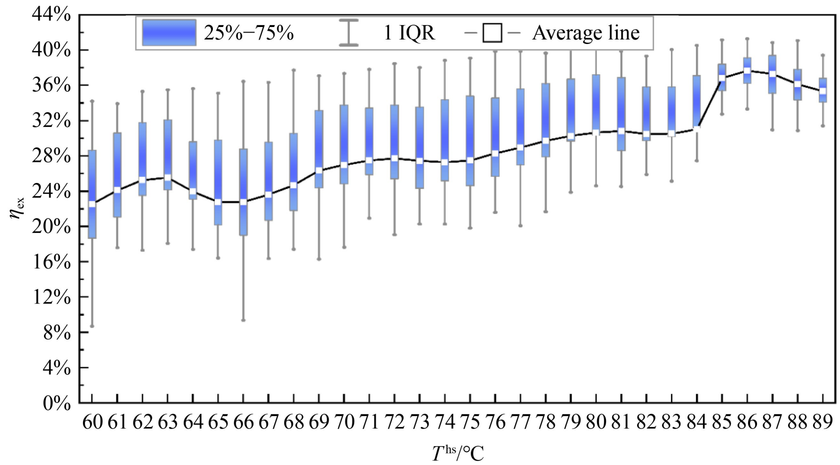

Fig. 9 Impact of heat source temperature on η_{ex}.

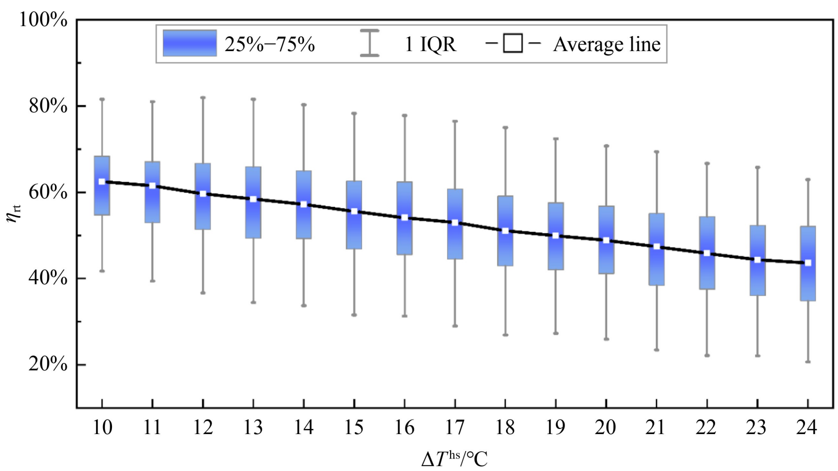

Fig. 10 Impact of heat source temperature difference on η_{rt}.

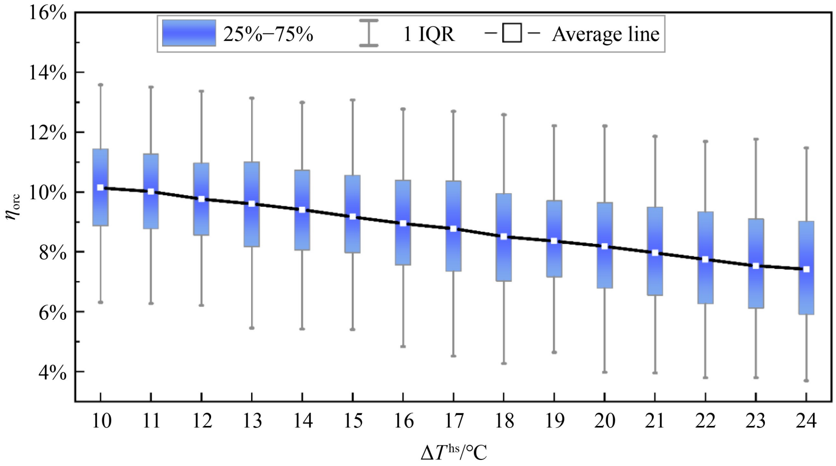

Fig. 11 Impact of heat source temperature difference on η_{orc}.

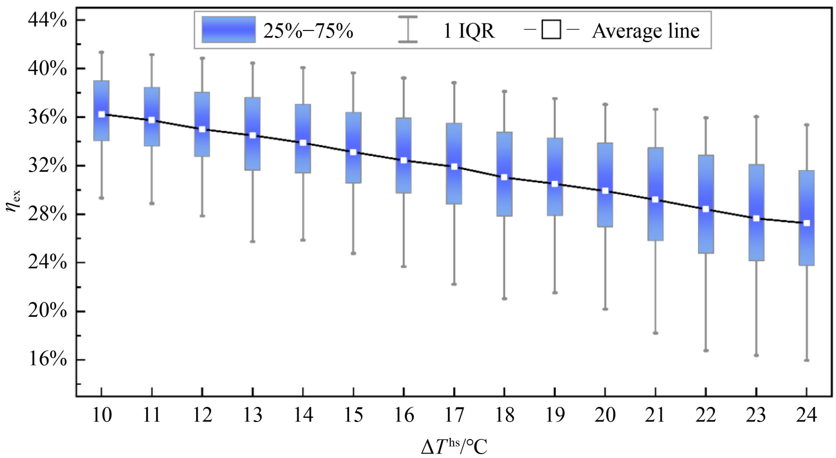

Fig. 12 Impact of heat source temperature difference on η_{ex}.

27.8%. This decline is attributed to the increased irreversible heat losses caused by the higher temperature difference, resulting in an overall decrease in the TI-CB performance. However, because the sensitivity of these indicators to temperature difference is relatively low, fluctuations within the sub-intervals are more pronounced, as reflected by the larger ranges in the box plots.

In addition, the influence of heat pump temperature rise on the overall system characteristics is investigated. The impact of the heat pump temperature difference on the main performance indicators of the TI-CB is shown in Figs. 13 and 14. As the heat pump temperature rise increases, the mean COP of the TI-CB decreases from 7.6 to 4.8, while the mean η_{orc} increases from 7.0% to 10.2%.

In TI-CB, the thermal storage temperature is largely constrained. To maintain a stable thermal storage temperature, the condensation temperature must remain high. An increase in the heat pump temperature rise lowers the evaporation temperature. To maintain a stable heat transfer rate in the evaporator, the mass flow rate of the working fluid must be reduced, which increases the pressure difference between the condenser and evaporator. This, in turn, results in higher compressor power consumption, leading to a decrease in COP.

The increase in η_{orc} is due to the decoupling of the ORC in the energy discharging phase from the energy charging phase. As the heat pump temperature rise increases, the condenser temperature also rises, improving the quality of the thermal energy transferred during both energy charging and discharging, resulting in an upward trend in η_{orc}.

4.4 Characteristic analysis of off-design condition parameters

When fluctuations in the actual industrial integrated energy system cause changes in the boundary conditions

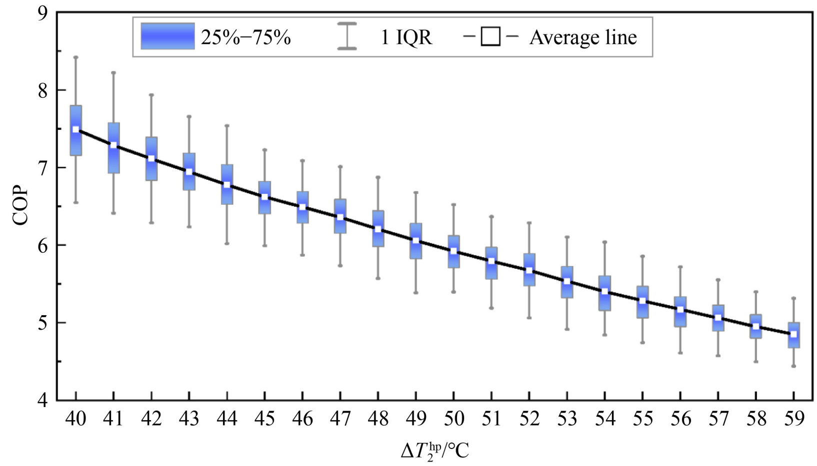

Fig. 13 Impact of heat pump temperature rise on COP.

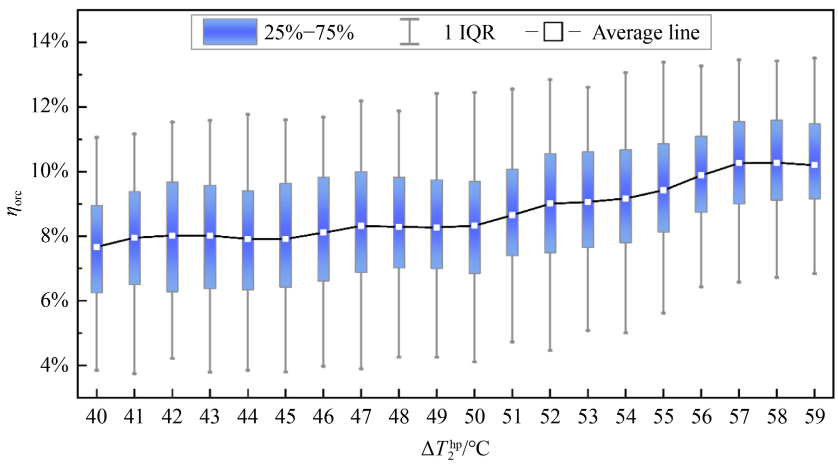

Fig. 14 Impact of heat pump temperature rise on η_{orc}.

of TI-CB, adjustments can be made by regulating the working fluid in both the HP and ORC during the energy charging and discharging phases to mitigate the impact of these disturbances. Additionally, variations in the mass flow rate of the working fluid can affect the performance of components such as the evaporator, condenser, compressor, and expander, which in turn influence the overall performance of TI-CB.

As an example, consider a set of TI-CB parameters within the feasible space. When the mass flow rates of the HP and ORC fluctuate within ±30% of their design conditions, the resulting changes in system performance are illustrated below.

Figures 15 and 16 show the variations in COP and η_{orc} under off-design conditions. As shown, COP remains largely unaffected by fluctuations in the ORC mass flow rate, while η_{orc} shows no significant variation with changes in the HP mass flow rate. Since the HP and ORC operate relatively independently, the performance of HP

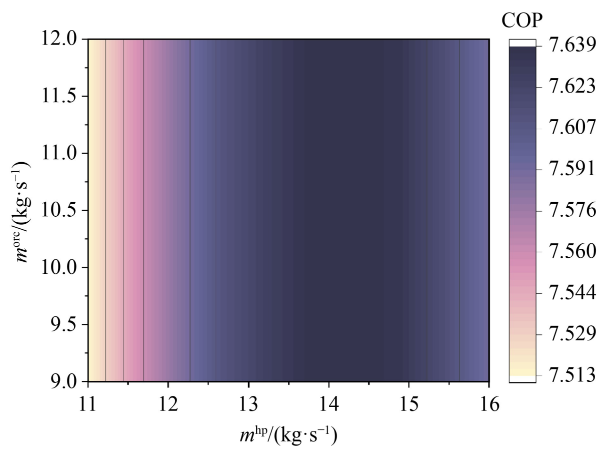

Fig. 15 Variation of COP under off-design conditions.

does not directly affect the ORC. Instead, it indirectly influences the ORC efficiency through the heat

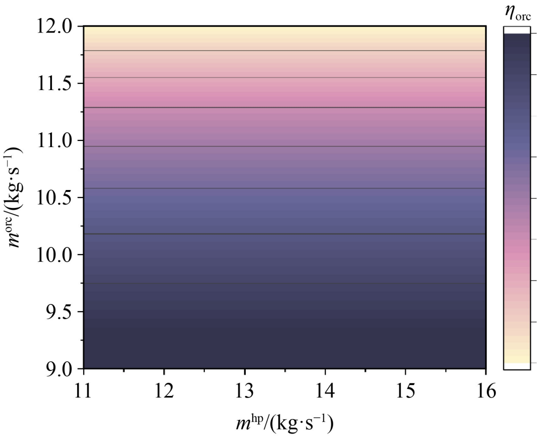

Fig. 16 Variation of η_{orc} under off-design conditions.

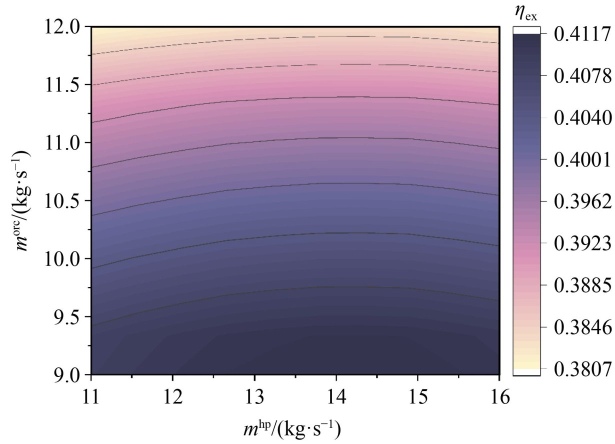

Fig. 18 Variation of η_{ex} under off-design conditions.

transferred to the thermal storage system and the resulting changes in the mass flow rate of the working fluid in the storage system. Similarly, the performance of the ORC influences the HP indirectly.

Figures 17 and 18 illustrate the variations in η_{rt} and η_{ex} under off-design conditions. It can be observed that both the round-trip efficiency and exergy efficiency are influenced by fluctuations in the mass flow rates of both the HP and ORC. Figure 17 exhibits a general trend of higher values in the upper-left region and lower values in the lower-right region. In contrast, Fig. 18 shows the opposite pattern, with lower values at the top and higher values toward the bottom. Both figures also display denser contour lines along the vertical axis, indicating that the fluctuations in the ORC mass flow rate have a more significant impact on both the round-trip efficiency and exergy efficiency than fluctuations in the HP mass flow rate.

To further quantify this effect, Sensitivity analysis was performed at the central operating point using the range

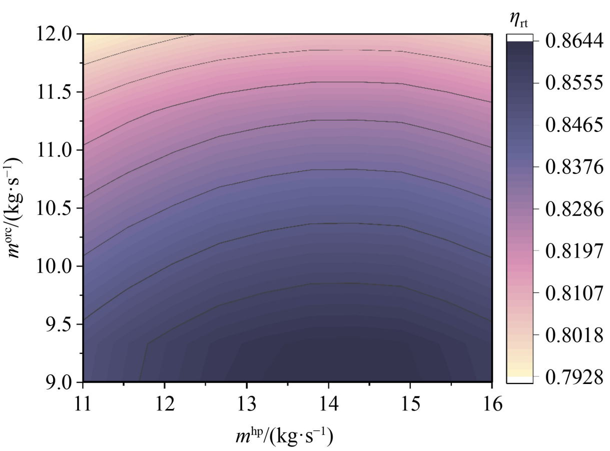

Fig. 17 Variation of η_{rt} under off-design conditions

analysis method. For η_{rt}, the variation induced by m^{orc} over its design range was 0.0582, whereas the variation caused by m^{hp} over its range was only 0.0138. For η_{ex}, this difference was even more pronounced: the variation induced by m^{orc} was 0.0285, while that caused by m^{hp} was only 0.0025. This is primarily because the expander characteristic curve in the ORC is more sensitive to changes in mass flow rate compared to the compressor characteristic curve in the HP.

4.5 Performance analysis of different working fluid parameters

Given the diversity of working fluid options in ORC systems, many researchers have investigated the impact of various working fluids on the performance of the ORC and Ranking-type Carnot battery [34–36]. This section extends the analysis beyond the baseline fluid R1233zd(E), by introducing three alternative fluids: R1234ze(E), R1224yd(Z), and R1336mzz(Z). The objective is to systematically evaluate the influence of different working fluids on the TI-CB's performance under design conditions.

Since the off-design condition analysis in this study primarily focuses on the qualitative influence of mass flow rate variations and does not provide a quantitative comparative basis for comparing different fluids, and considering that fluid property differences introduce complex coupling effects under off-design scenarios, the current comparison of working fluids is limited to design-point conditions. This approach provides a theoretical foundation for future optimization of working fluids under off-design operation.

The four working fluids exhibit significant differences in their effects on system performance, which are shown in Fig. 19. R1336mzz(Z) shows the best performance for T_{in}^{hs}, ΔT^{hs} and ΔT_2^{hp} under most conditions, primarily due

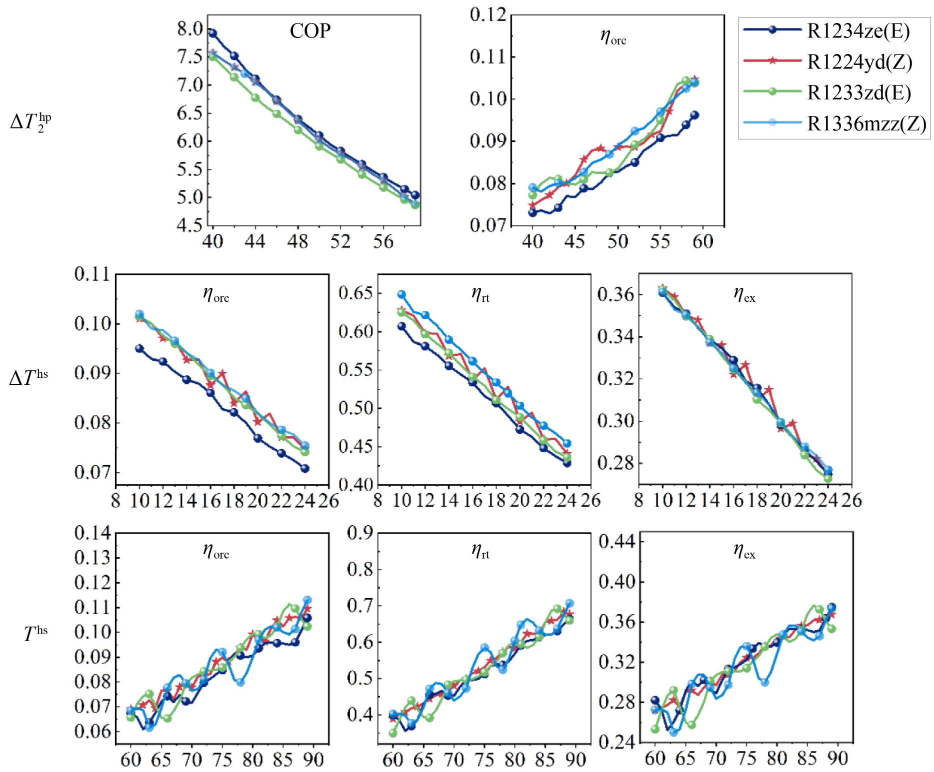

Fig. 19 Design condition parameter analysis under multiple working fluids.

to its unique molecular structure. As a hydrofluorolefin with a double bond, R1336mzz(Z) has favorable saturated vapor pressure characteristics and latent heat of phase change in the medium-to-high temperature range. This enables a higher enthalpy difference in the ORC cycle, thus improving power generation efficiency. However, its performance curve shows significant fluctuations, especially with changes in T_{in}^{hs}, which exhibit strong nonlinear behavior. This is closely related to its relatively low critical point and the sensitivity of its thermophysical properties to temperature.

R1233zd(E), used as the reference fluid, demonstrates good stability across the entire operating range, with a relatively smooth performance curve and minimal fluctuations. This stability is attributed to its symmetric molecular structure and higher critical temperature, resulting in milder thermodynamic performance variations within the industrial waste heat temperature range. Although its peak efficiency is slightly lower than that of R1336mzz(Z), its predictability and stability provide clear advantages for practical engineering applications.

The performance characteristics of R1234ze(E) and R1224yd(Z) are relatively similar, with subtle differences. R1234ze(E) performs more stably at low temperatures, while R1224yd(Z) shows superior performance at higher temperatures. This difference reflects the distinct molecular structures of the two fluids: R1234ze(E), with a trans structure, offers better low-temperature stability, whereas R1224yd(Z), with a cis structure, exhibits superior thermodynamic properties at higher temperatures.

It is noteworthy that R1224yd(Z) exhibits significant thermal slip. On one hand, this temperature slip improves the temperature match between the working fluid and the heat source, reducing irreversible heat transfer losses. On the other hand, the non-isothermal phase change increases the design complexity of the heat exchanger. As a zeotropic working fluid, R1224yd(Z) shows good thermodynamic performance under certain conditions, its thermal slip characteristics require careful evaluation in practical engineering applications, balancing the trade-off between design complexity and performance improvement. This is because it is a zeotropic working fluid, whose constituent components possess different boiling points. This causes its temperature to change continuously during a constant-pressure phase change rather than remaining constant. The engineering complexity arising from this non-isothermal phase change is that it renders traditional design methods, such as pinch point analysis and the logarithmic mean temperature difference, ineffective. Consequently, more sophisticated segmented numerical models must be employed, along with high-efficiency pure counter-flow heat exchanger configurations, to ensure design accuracy and performance. While R1224yd(Z) can achieve good thermodynamic performance under certain conditions, its thermal slip characteristics require careful consideration in practical engineering applications, balancing design complexity against potential performance gains.

5 Conclusions

This paper established a quasi-dynamic mathematical model for a TI-CB system, targeting industrial waste heat utilization in the 60−90 °C range and incorporating both design and off-design conditions. It constructed a multi-operating condition set to conduct performance analyses, including parameter sensitivity, design-condition, and off-design condition studies. The main conclusions are as follows:

1) Parameter sensitivity analysis and design-condition analysis jointly reveal that the heat source utilization parameters ($T_{\rm in}^{\rm hs}$ and $\Delta T^{\rm hs}$) and the heat pump temperature lift ($\Delta T_2^{\rm hp}$) are the dominant factors affecting TI-CB performance. Increasing the heat source temperature and reducing the heat source-side temperature difference significantly improve system efficiency, while the heat pump temperature lift introduces a trade-off between charging COP and discharging $\eta_{\rm orc}$.

2) Design condition analysis demonstrates substantial performance improvements with increasing heat source temperature. When the heat source temperature rises from 60 to 90 °C, $\eta_{\rm orc}$ increases from 6.8% to 10.1%, $\eta_{\rm rt}$ from 34.3% to 67.9%, and $\eta_{\rm ex}$ from 25.3% to 35.4%. However, larger heat source temperature differences reduce system efficiency due to increased irreversible heat losses. Heat pump temperature rise shows contrasting effects on COP and $\eta_{\rm orc}$: COP decreases from 7.6 to 4.8 while $\eta_{\rm orc}$ increases from 7.0% to 10.2% as the temperature rise increases.

3) Off-design condition analysis demonstrates that fluctuations in the ORC mass flow rate have a far more significant impact on overall system efficiency ($\eta_{\rm rt}$ and $\eta_{\rm ex}$) than in the HP mass flow rate. This is primarily attributed to the higher sensitivity of the expander's characteristic curve in the ORC to mass flow variations compared to the compressor in the HP , highlighting the importance of ORC-side control strategies in practical operation.

4) Working fluid comparison indicates that R1336mzz(Z) exhibits superior thermodynamic performance, although its performance curve shows significant nonlinear fluctuations. In contrast, R1233zd(E) demonstrates the best stability across the entire operating range, making it more suitable for practical engineering applications that prioritize reliable and predictable operation.

Based on the current study, future work will employ AI techniques to train multi-condition mapping models for the Carnot battery, improving computational efficiency. System optimization will also focus on the addition of recuperators and selection of working fluids with improved thermal matching at each stage, aiming to exploit a wider and more diverse range of industrial waste heat resources.

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Competing Interests The authors declare no competing interests.

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